



AKTU

B.Tech I-Year

Electrical Engg.

ONE SHOT Revision

Unit-1

DC Circuits



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AKTU B.Tech First Year

Fundamentals of Electrical Engineering

Unit-1 : DC Circuits

AKTU Syllabus

(Electrical circuit elements (R, L and C), Concept of active and passive elements,
(Voltage and current sources, (concept of linearity,) (unilateral and bilateral
elements.)
Kirchhoff's laws, Mesh and nodal methods of analysis.

① **Active Elements:-** Elements that can provide energy to the networks.

Examples: Voltage Source & Current Source , Battery

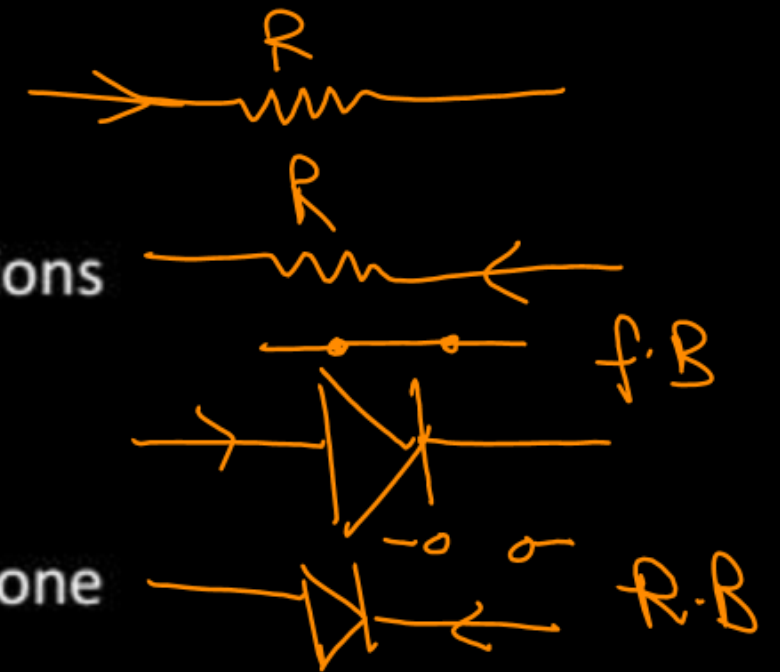
✓ **Passive Elements:-** Elements that can which can only receive energy, which it can either dissipate, absorb or store it in an electric field or a magnetic field.

Examples: R, L, C , Resistance - dissipate , Inductor & Capacitor → Energy store

② **Bilateral Element & Unilateral Element**

Bilateral Element:- Elements which allows current to flow from both directions are known as Bilateral Element. (**Examples: Resistance**)

Unilateral Element:- Elements which allows current to flow only from one direction are known as Unilateral Element . (**Examples: Diode**)



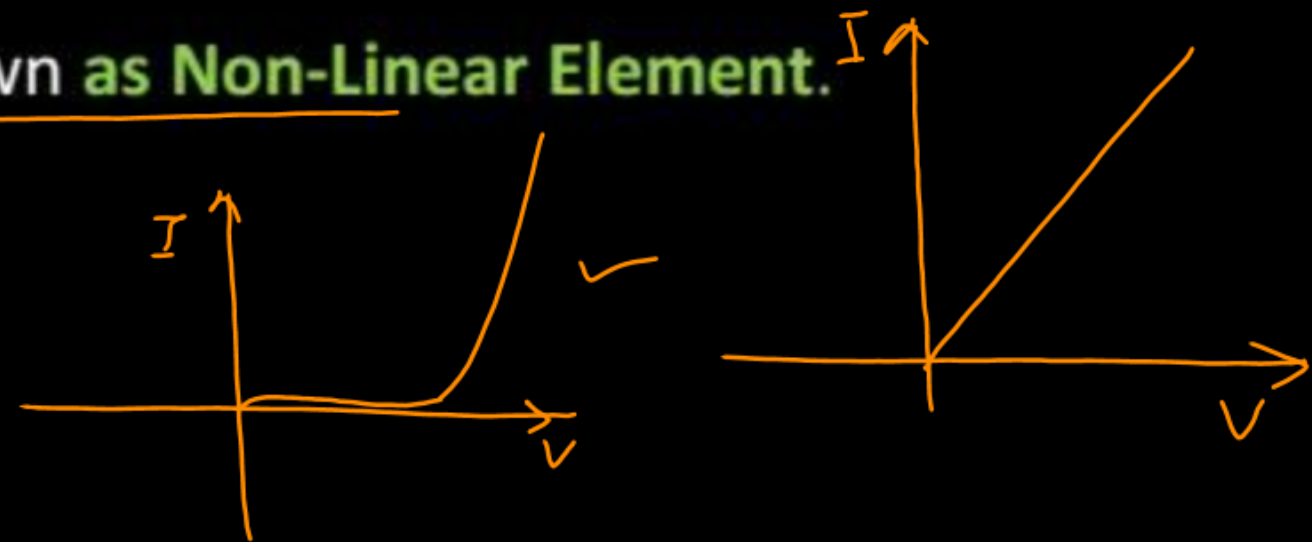
Linear and Non Linear Elements

- ③ **Linear & Non Linear Element** – An Element is said to be **linear**, if it obeys the principle of homogeneity and superposition, Otherwise it is known as **Non-Linear Element**.

Examples: Resistance (Linear), diodes (Non-Linear)

↓
Semiconductor ↗

Lumped and Distributed network



- ! **Lumped Network**:- A network in which all the network elements are physically separable is known as a lumped network.

Examples: - Network having Resistance, Inductor or Capacitor etc

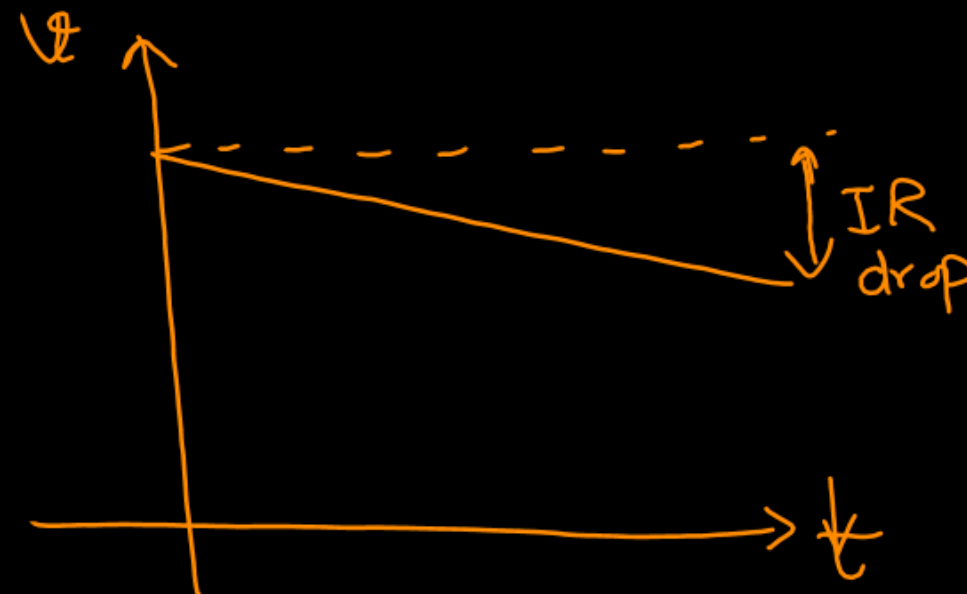
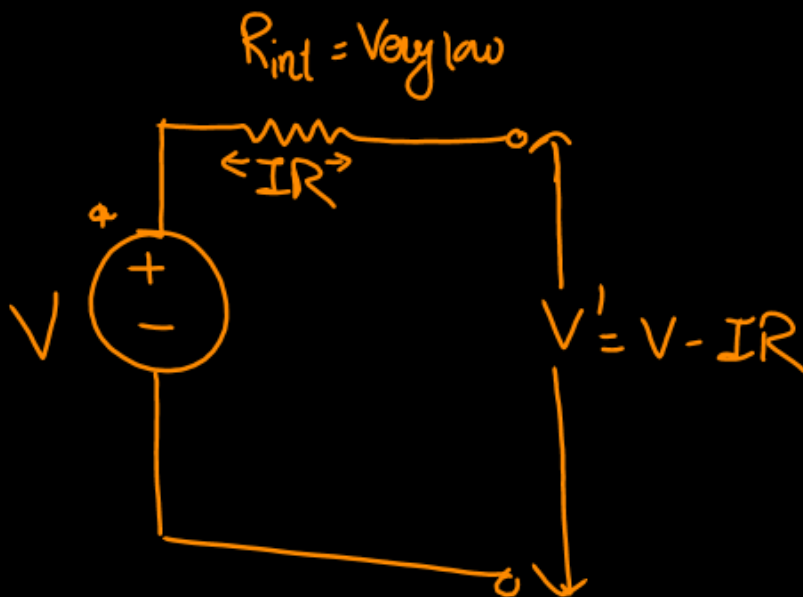
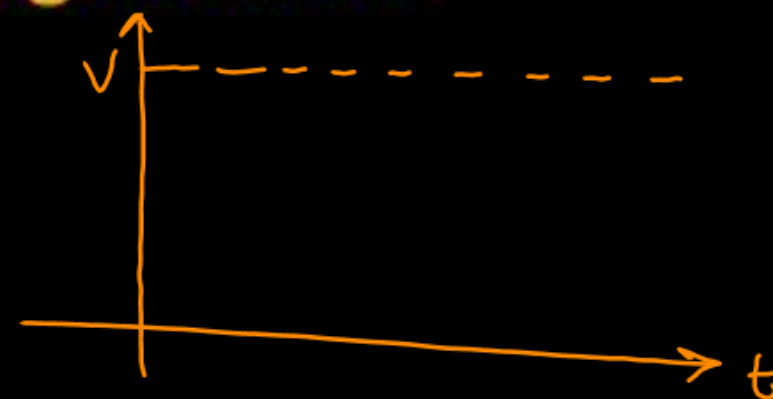
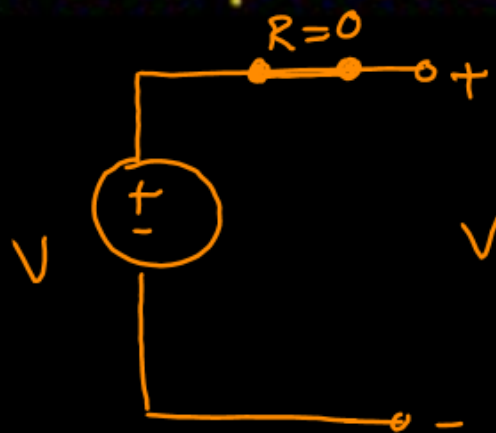
- ✓ **Distributed Network** – A network in which network elements can not be physically separable is known as **Distributed network** like Transmission Line.

Voltage Source: Ideal and practical Voltage source

Ideal Voltage source:- Energy Source which provide constant voltage across its terminals irrespective of current drawn through its terminals.

It has Zero Internal Resistance

Practical Voltage Source:- A practical voltage source has small internal resistance due to which voltage across terminals decrease slightly with increase in current.

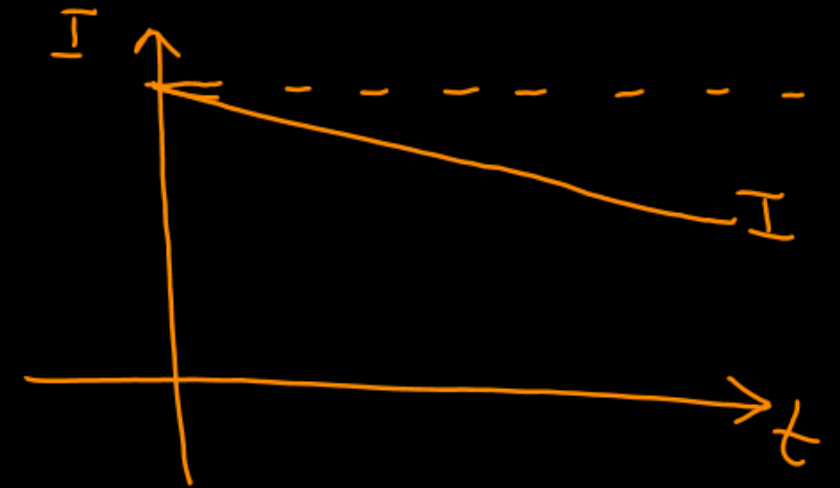
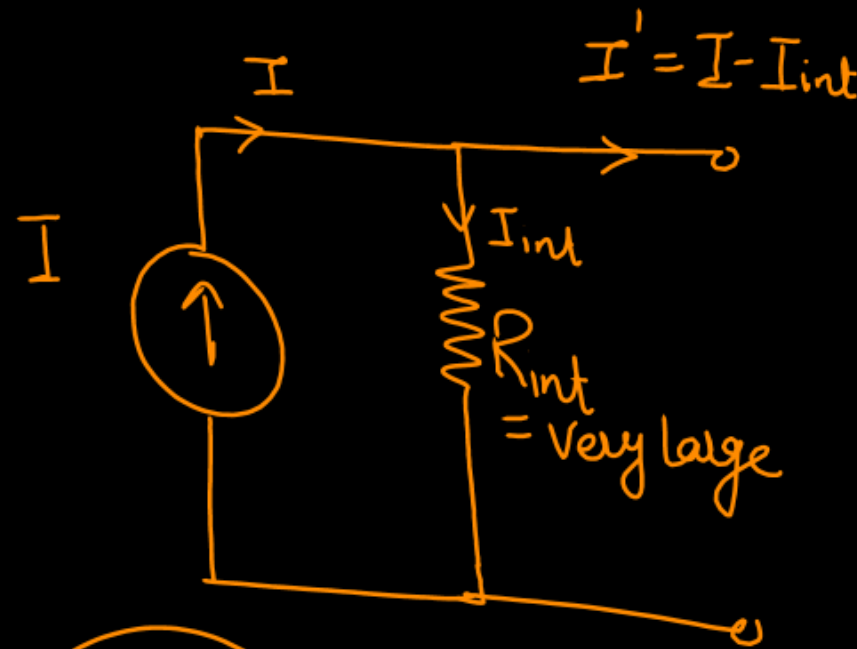
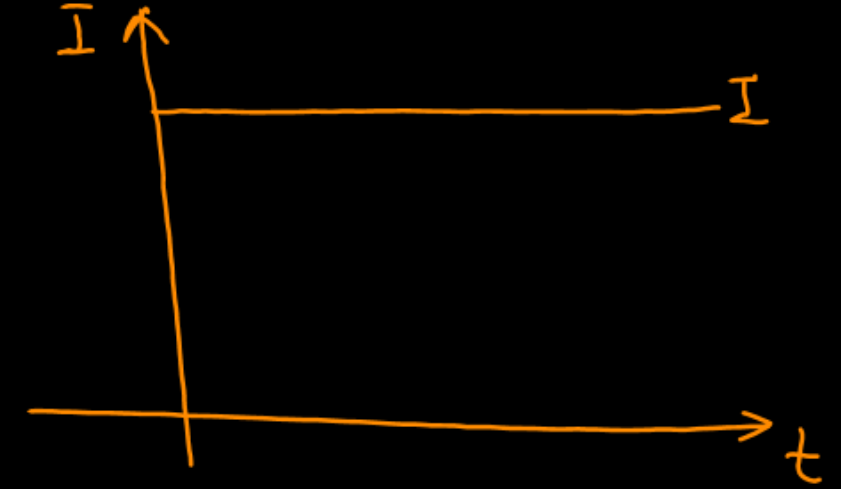
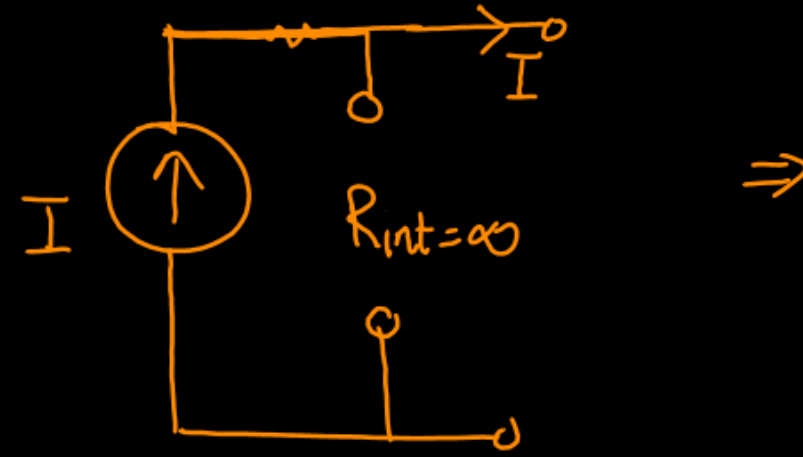


Current Source: Ideal and Practical Current source

Ideal Current source:- Energy Source which provide constant Current across its terminals irrespective of current drawn through its terminals.

It has infinite Internal Resistance.

Practical Current Source:- A practical Current source has very high internal resistance due to which Current across terminals decrease slightly with increase in voltage.



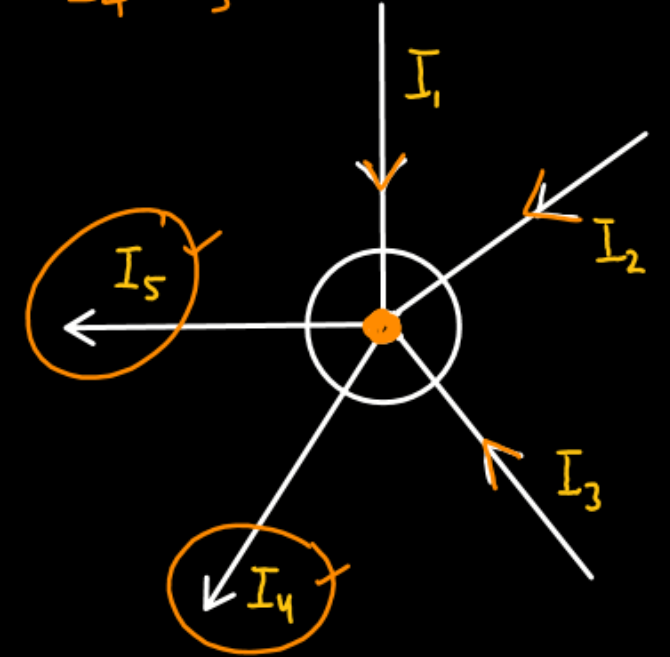
$$R = \infty$$
$$V = IR = \infty$$

$$I_1 + I_2 + I_3 = I_4 + I_5$$

Kirchhoff's Current Law:- According to Kirchhoff's Current Law, algebraic sum of ALL the currents entering and leaving a node must be equal to zero, $I_{(\text{exiting})} + I_{(\text{entering})} = 0$.

Nodal Analysis

$$(I_4 + I_5) - I_1 - I_2 - I_3 = 0$$



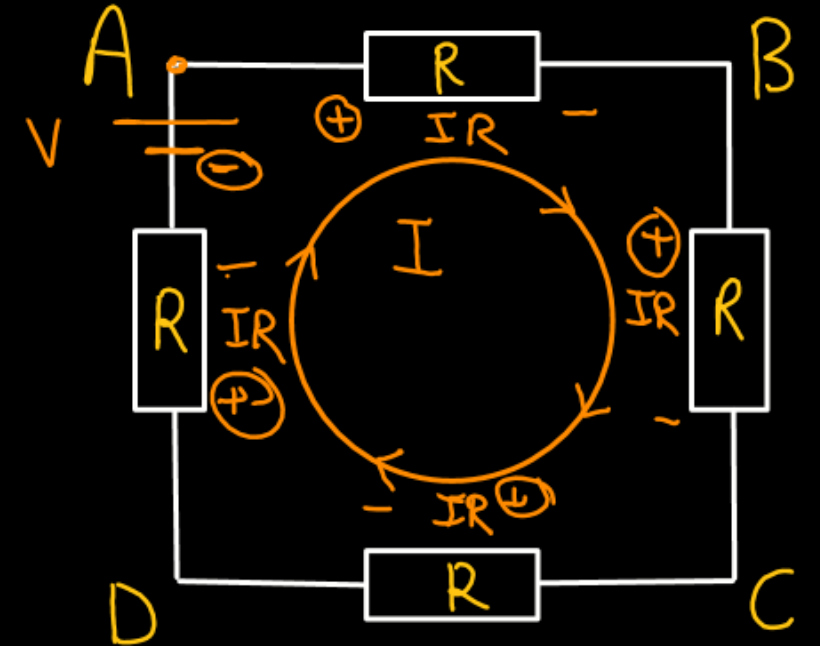
Kirchhoff's Voltage Law:- According to Kirchhoff's Voltage Law, The voltage around a loop equals the sum of every voltage drop in the same loop for any closed network and equals zero.

Mesh Analysis

Why We can not apply KVL in a loop having Current Source.

Ans. NO

$$IR + IR + IR + IR - V = 0$$



Ques. Explain the procedure of mesh analysis with the help of an example

(AKTU-2023-24)

Step-1 - Identify loops in given circuit

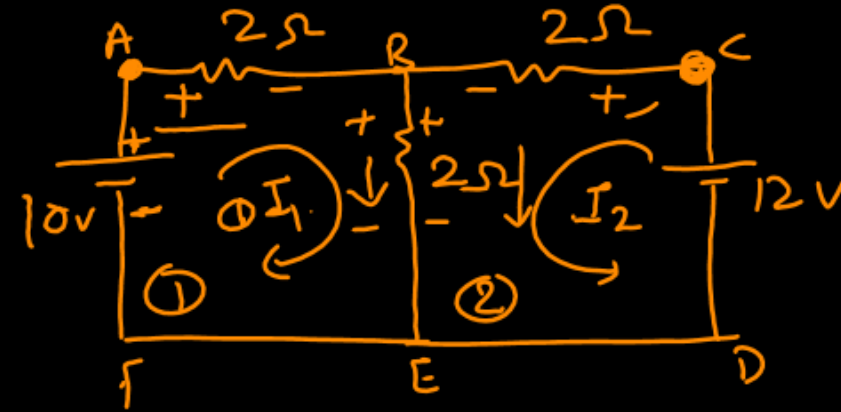
Step-2 Consider loop current in mesh (I_1, I_2)

Step-3 (optional) $\rightarrow$ mark polarity on each resistance

Step-4 Apply KVL in each ~~loop~~ mesh one by one

Step-5 - Now we have one equation from each mesh

Step-6 Solve all equations to find mesh currents.



Apply KVL in mesh ①

$$+2I_1 + 2(I_1 + I_2) - 10 = 0$$

$$4I_1 + 2I_2 = 10 \quad \text{--- ①}$$

Apply KVL in mesh ②

$$+2I_2 + 2(I_2 + I_1) = 12$$

$$2I_1 + 4I_2 = 12 \quad \text{--- ②}$$

$$I_1 = 1.33 \text{ A}$$

$$I_2 = 2.33 \text{ A}$$

Loop $\rightarrow$ Any closed path

Mesh $\rightarrow$ shortest closed path loop that does not contain any other loop

Ques. Explain the procedure of Nodal analysis with the help of an example.

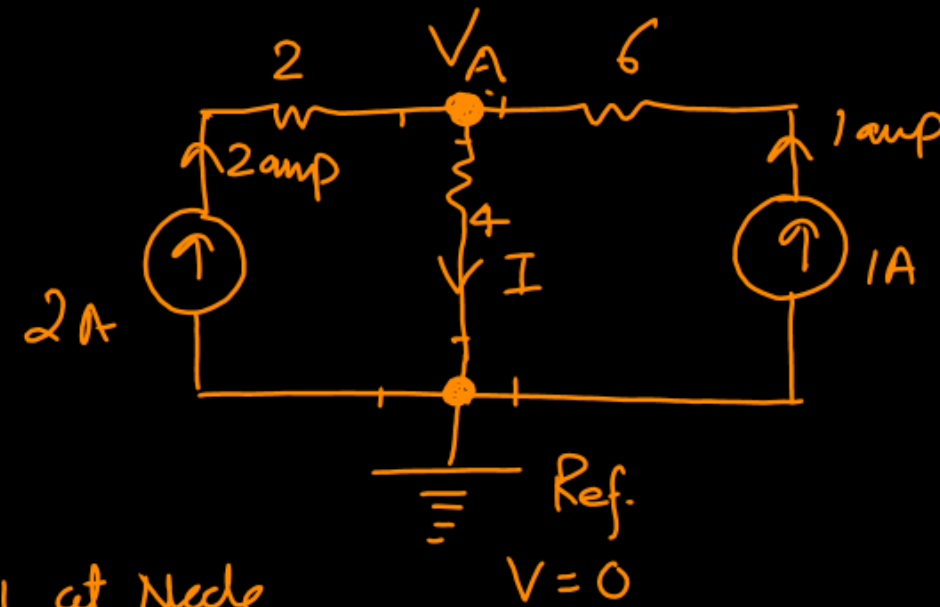
Step 1 → Identify junction in the circuit

Step 2 → Consider one junction at reference ($V=0$)
and rest junction voltages should be considered

Step 3 Now assume ~~random~~ branch current with random direction

Step 4 Now apply KCL at each node

Step 5 Solve these equations to find Node Voltages and Currents.



Apply KCL at Node

$$1 + 2 = I$$

$$\boxed{I = 3} \text{ amp}$$

$$I = \frac{V_A - 0}{4} = \frac{V_A}{4}$$

$$\underline{V_A = 4I = 4 \times 3 = 12 \text{ Volt}}$$

Junction - where 3 or more branches meet

Node → where 2 or more branches meet

Q.1 Using Mesh Analysis Find Current in all branches.

Apply KVL in Supermesh.

$$+5I + 2(I+5) + 6(I+5+I_1) - 60 = 0$$

$$5I + 2I + 10 + 6I + 30 + 6I_1 - 60 = 0$$

$$13I + 6I_1 = 20 \quad \text{--- (1)}$$

Apply KVL in loop (2)

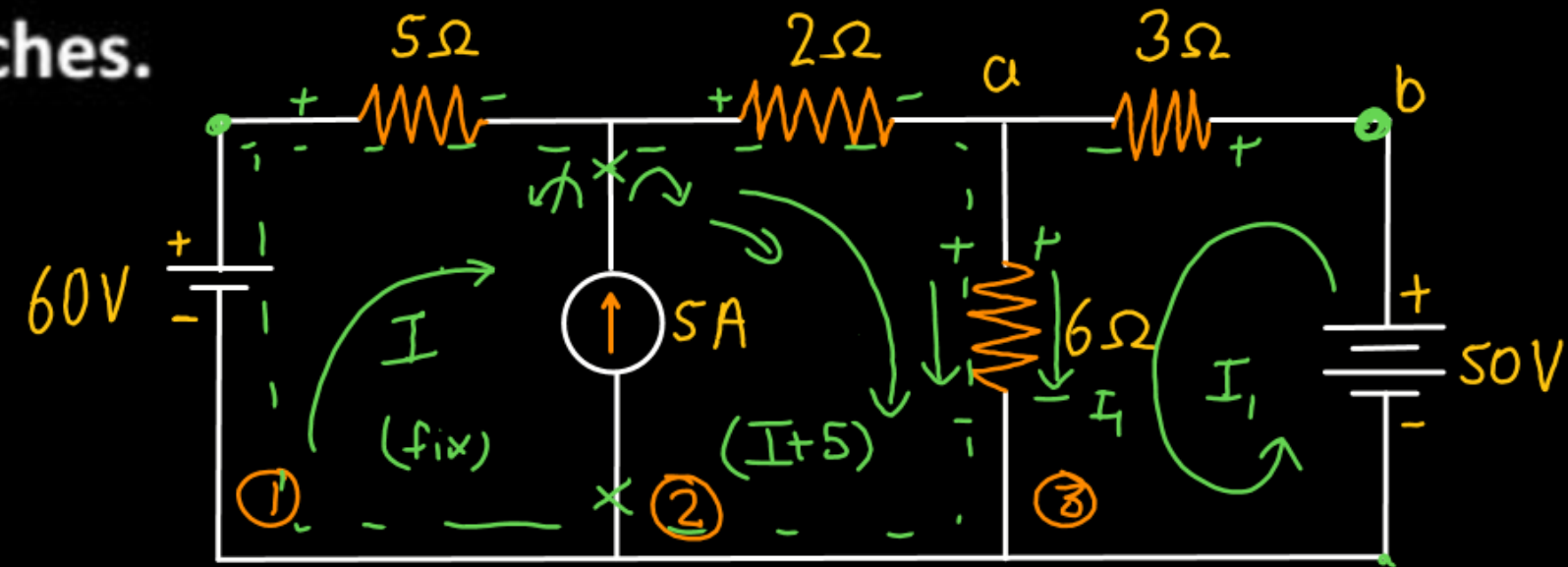
$$+3(I_1) + 6(I_1 + I + 5) - 50 = 0$$

$$3I_1 + 6I_1 + 6I + 30 - 50 = 0$$

$$6I + 9I_1 = 20 \quad \text{--- (2)}$$

Solve Equ (1) & (2)

$$\begin{aligned} I &= 0.74 \text{ Amp} \\ I_1 &= 1.728 \text{ Amp} \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \underline{\text{Ans}}$$



Q.2 Using Mesh Analysis Find Current in $20\ \Omega$ resistance. (AKTU-2023-24)

Apply KVL in Loop ①

$$5I_1 + 10(I_1 - I_2) = 20$$

$$15I_1 - 10I_2 = 20 \quad \text{--- ①}$$

Apply KVL in Loop ②

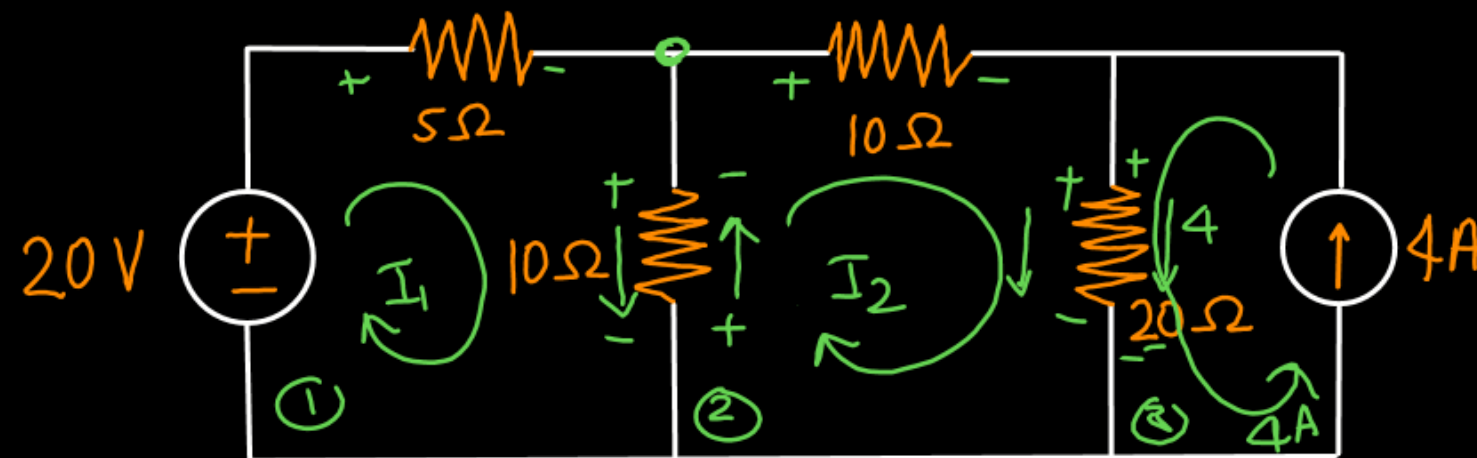
$$+10(I_2) + 20(I_2 + 4) + 10(I_2 - I_1) = 0$$

$$-10I_1 + 40I_2 = -80 \quad \text{--- ②}$$

Solve Equ ① & ②,

$$I_1 = 0$$

$$I_2 = -2 \text{ Amp}$$



$$I_{20\Omega} = I_2 + 4$$

$$= -2 + 4$$

$$I_{20\Omega} = 2 \text{ Amp}$$

Ans

Mesh Analysis(KVL): Numerical Problems

Q.3 Using Mesh Analysis Find Current in 6 ohm Resistance.

Apply KVL in Loop ①

$$+3I_1 + 6(I_1 - I_2) = 6$$

$$9I_1 - 6I_2 = 6 \quad \text{--- ①}$$

Apply KVL in Loop (2 & 3) (supermesh)

$$2(I_2) + 1(I_2 + 3) + 4(I_2 + 3) + 6(I_2 - I_1) = 0$$

$$-6I_1 + 13I_2 + 15 = 0$$

$$-6I_1 + 13I_2 = -15 \quad \text{--- ②}$$

Solve Eqn ① & ②

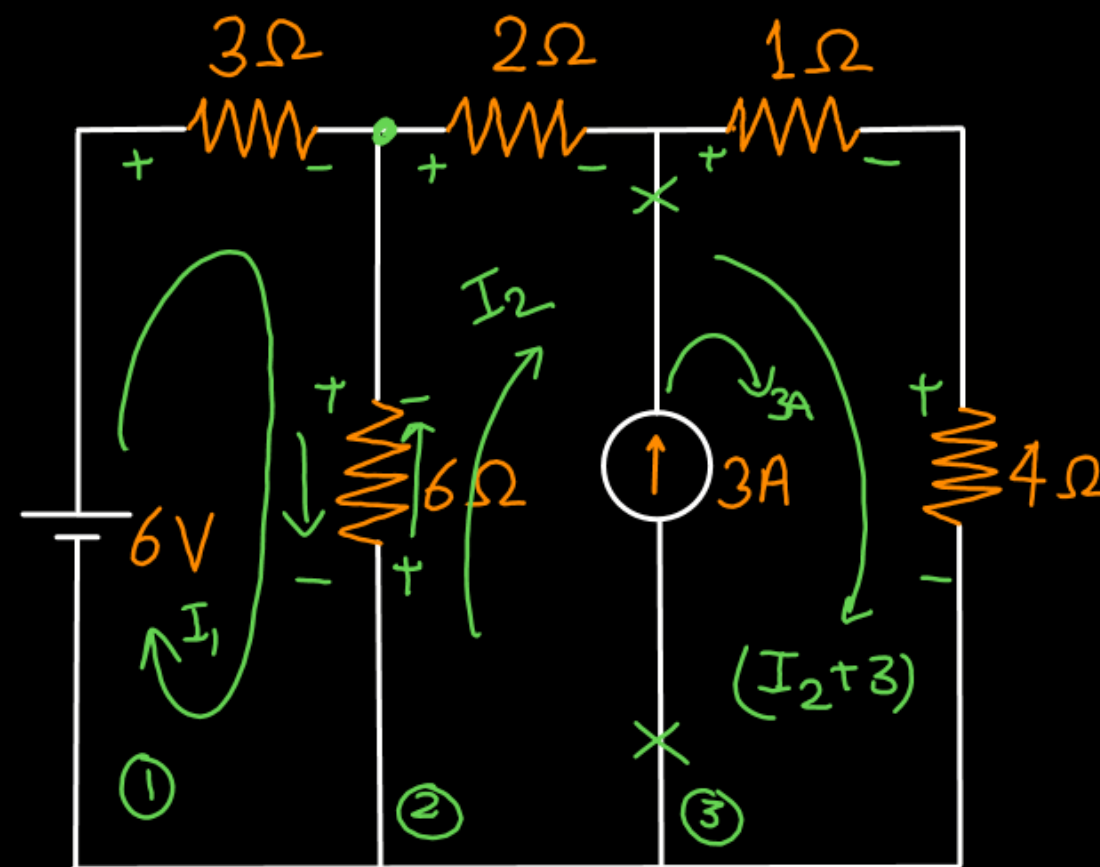
$$I_1 = -0.148 \text{ A}$$

$$I_2 = -1.22 \text{ Amp}$$

$$I_{6\Omega} = (I_1 - I_2)$$

$$= -0.148 + 1.22$$

$$I_{6\Omega} = 1.072 \downarrow$$



using Mesh current method? (AKTU 2023-24)

Apply KVL in Loop ①

$$+30(I_1 - I_3) + 50(I_1 - I_2) + 20 - 100 + 60(I_1) = 0$$

$$140I_1 - 50I_2 - 30I_3 = 80 \quad \text{--- ①}$$

Apply KVL in Loop ②

$$+40(I_2 - I_3) + 10(I_2) - 50 - 20 + 50(I_2 - I_1) = 0$$

$$-50I_1 + 100I_2 - 40I_3 = 70 \quad \text{--- ②}$$

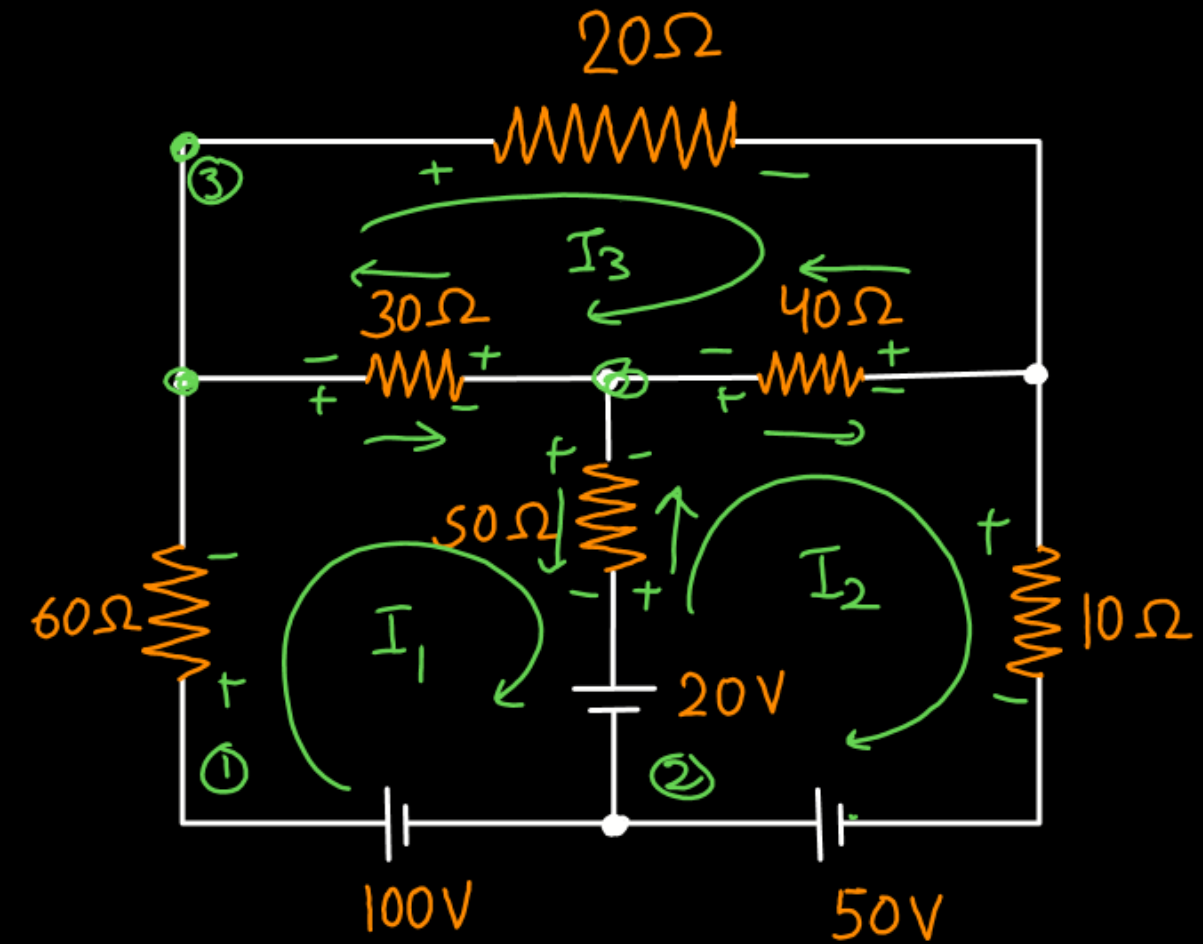
Apply KVL in Loop ③

$$+20I_3 + 40(I_3 - I_2) + 30(I_3 - I_1) = 0$$

$$-30I_1 - 40I_2 + 90I_3 = 0 \quad \text{--- ③}$$

Solve eqs ①, ② & ③

$$I_1 = 1.65 \text{ A}, I_2 = 2.12 \text{ A}, I_3 = 1.49 \text{ A}$$



Q.5 Determine the set of Mesh equations that are required to solve the network given in below circuit diagram. **AKTU 2022-23**

Apply KVL in Loop ① $I_3 = -3A$

$$+2I - 12 + 5(I_1 + 3) = 0$$

$$2I + 5I_1 = 12 - 15 \quad \text{--- ①}$$

Apply KVL in Loop ②

$$+6 - 2I + 6(I_2 + 3) = 0$$

$$6 - 2I + 6I_2 + 18 = 0$$

$$-2I + 6I_2 = -24 \quad \text{--- ②}$$

Now we have

$$I = -3 - I_2$$

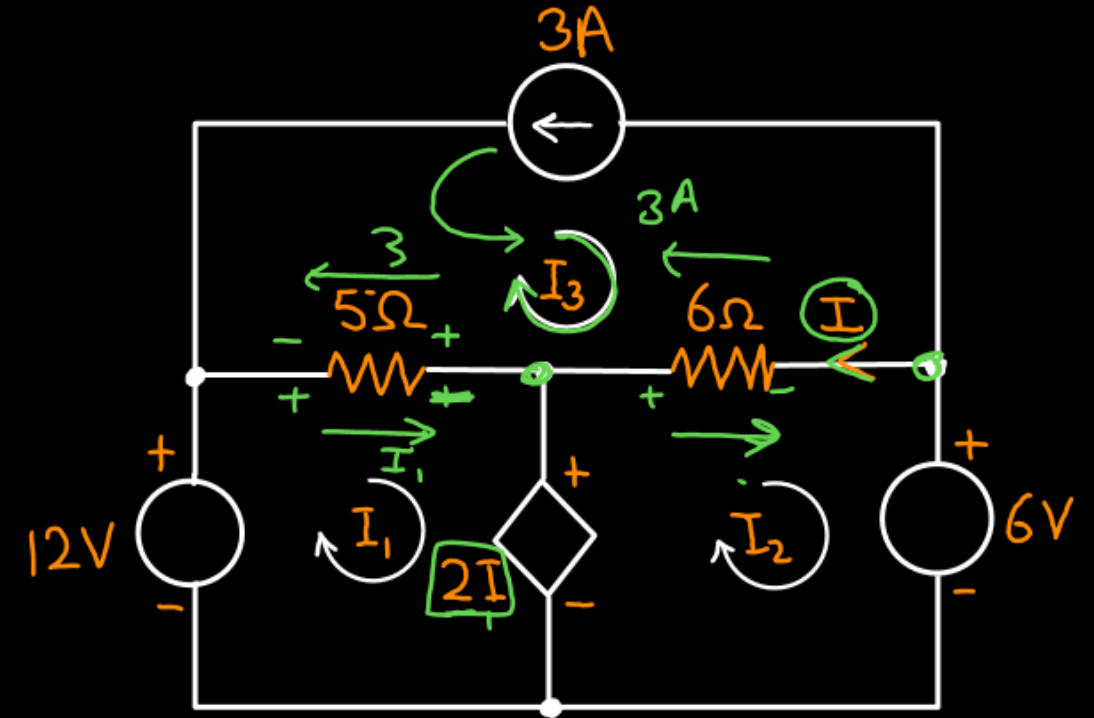
$$I + 0I_1 + I_2 = 3 \quad \text{--- ③}$$

$$I = 0.75$$

$$I_1 = -0.9 A$$

$$I_2 = -3.75 A$$

Ans



Q.6 Use nodal analysis to find the currents in various resistors of the circuit shown below.

$$i_1^o = \frac{V_A - 0}{2} = \frac{V_A}{2}, \quad i_2^o = \frac{V_A - V_C}{5}, \quad i_3^o = \frac{V_A - V_B}{3}, \quad i_4^o = \frac{V_B}{5}, \quad i_5^o = \frac{(V_B - V_C)}{1}, \quad i_6^o = \frac{V_C}{4} = \frac{3.8}{4} = 0.95$$

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Apply KCL at Node (A)

$$i_1^o + i_2^o + i_3^o = 10$$

$$\frac{V_A}{2} + \frac{V_A - V_C}{5} + \frac{V_A - V_B}{3} = 10$$

$$V_A \left(\frac{1}{2} + \frac{1}{5} + \frac{1}{3} \right) - \frac{V_B}{3} - \frac{V_C}{5} = 10$$

$$V_A \left(\frac{31}{30} \right) - \frac{V_B}{3} - \frac{V_C}{5} = 10 \quad \text{--- (1)}$$

Apply KCL at Node (B)

$$i_3^o = i_4^o + i_5^o$$

$$i_3^o - i_4^o - i_5^o = 0$$

$$\frac{V_A - V_B}{3} - \frac{V_B}{5} - V_B + V_C = 0$$

$$\frac{V_A}{3} - V_B \left(\frac{1}{3} + \frac{1}{5} + 1 \right) + V_C = 0$$

$$\frac{V_A}{3} - V_B \left(\frac{23}{15} \right) + V_C = 0 \quad \text{--- (2)}$$

Apply KCL at Node (C)

$$i_2^o + i_5^o = i_6^o + 2$$

$$i_2^o + i_5^o - i_6^o = 2$$

$$\frac{V_A - V_C}{5} + V_B - V_C - \frac{V_C}{4} = 2$$

$$\frac{V_A}{5} + V_B - V_C \left(\frac{1}{5} + 1 + \frac{1}{4} \right) = 2$$

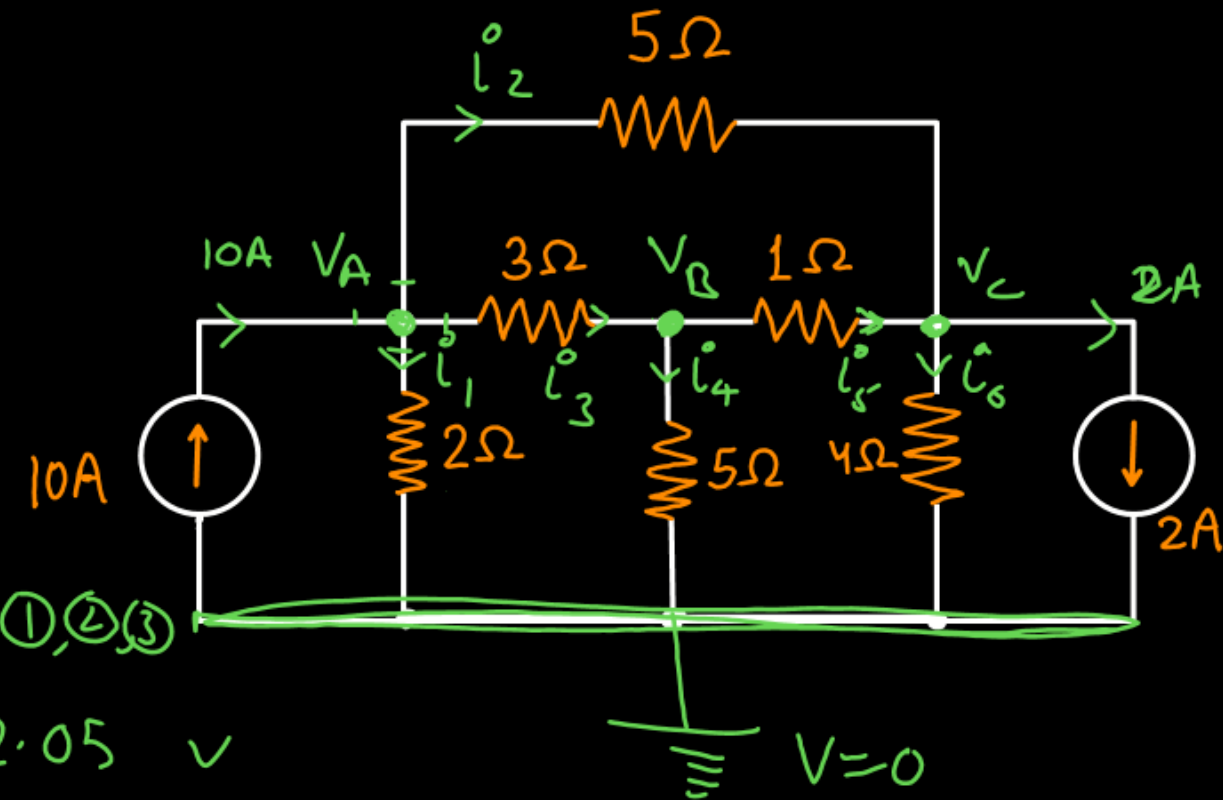
$$\frac{V_A}{5} + V_B - V_C \left(\frac{29}{20} \right) = 2 \quad \text{--- (3)}$$

Solve Eqn (1), (2), (3)

$$V_A = 12.05 \text{ V}$$

$$V_B = 5.1 \text{ V}$$

$$V_C = 3.8 \text{ V}$$



Q.7 Determine the currents in the various branches of the circuit shown in Figure by nodal analysis? (AKTU 2022-23)

$$I_1 = \frac{0 - V_B + 100}{20} = \frac{100 - V_B}{20}, \quad I_2 = \frac{V_B}{10}, \quad I_3 = \frac{V_B - V_C}{15}, \quad I_4 = \frac{V_C}{10}$$

$$I_5 = \frac{V_C - 80}{10}$$

Apply KCL at Node B

$$I_1 = I_2 + I_3$$

$$I_1 - I_2 - I_3 = 0$$

$$\frac{100 - V_B}{20} - \frac{V_B}{10} - \frac{V_B - V_C}{15} = 0$$

$$\frac{100}{20} - V_B \left(\frac{1}{20} + \frac{1}{10} + \frac{1}{15} \right) + \frac{V_C}{15} = 0$$

$$-V_B \left(-\frac{13}{60} \right) + \frac{V_C}{15} = -5 \quad \text{--- (1)}$$

Apply KCL at Node C

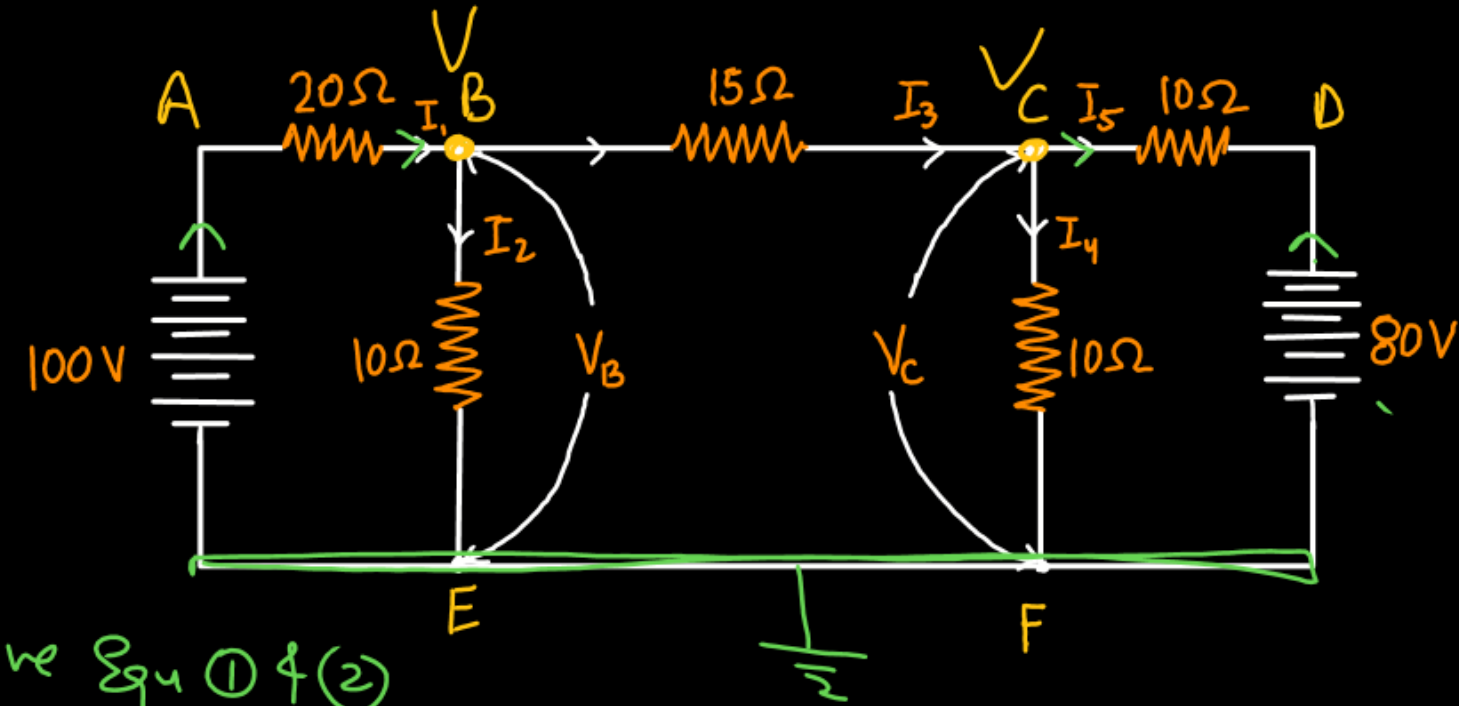
$$I_3 = I_4 + I_5$$

$$I_3 - I_4 - I_5 = 0$$

$$\frac{V_B - V_C}{15} - \frac{V_C}{10} - \frac{V_C - 80}{10} = 0$$

$$\frac{V_B}{15} - V_C \left(\frac{1}{15} + \frac{1}{10} + \frac{1}{10} \right) = -8$$

$$\frac{V_B}{15} - V_C \left(\frac{8}{30} \right) = -8 \quad \text{--- (2)}$$



Solve Equ (1) & (2)

$$V_B = 35 \text{ V}$$

$$V_C = 38.75 \text{ V}$$

Q.8 Determine the current by Nodal method, through 2 ohm resistor for the network shown below? (AKTU 2023-24)

$$I_1 = \frac{20 - V_A}{1}, \quad I_2 = V_A, \quad I_3 = \frac{(V_A - V_B)}{0.5} = 2(V_A - V_B), \quad I_4 = \frac{V_B}{2}, \quad I_5 = (20 - V_B)$$

KCL at (A)

$$I_1 = I_2 + I_3$$

$$20 - V_A = V_A + 2V_A - 2V_B$$

$$20 = 4V_A - 2V_B \rightarrow \textcircled{1}$$

KCL at Nod (B)

$$I_3 + I_5 = I_4$$

$$2V_A - 2V_B + 20 - V_B = 0.5V_B$$

$$2V_A - 3V_B - 0.5V_B = -20$$

$$2V_A - 3.5V_B = -20 \rightarrow \textcircled{2}$$

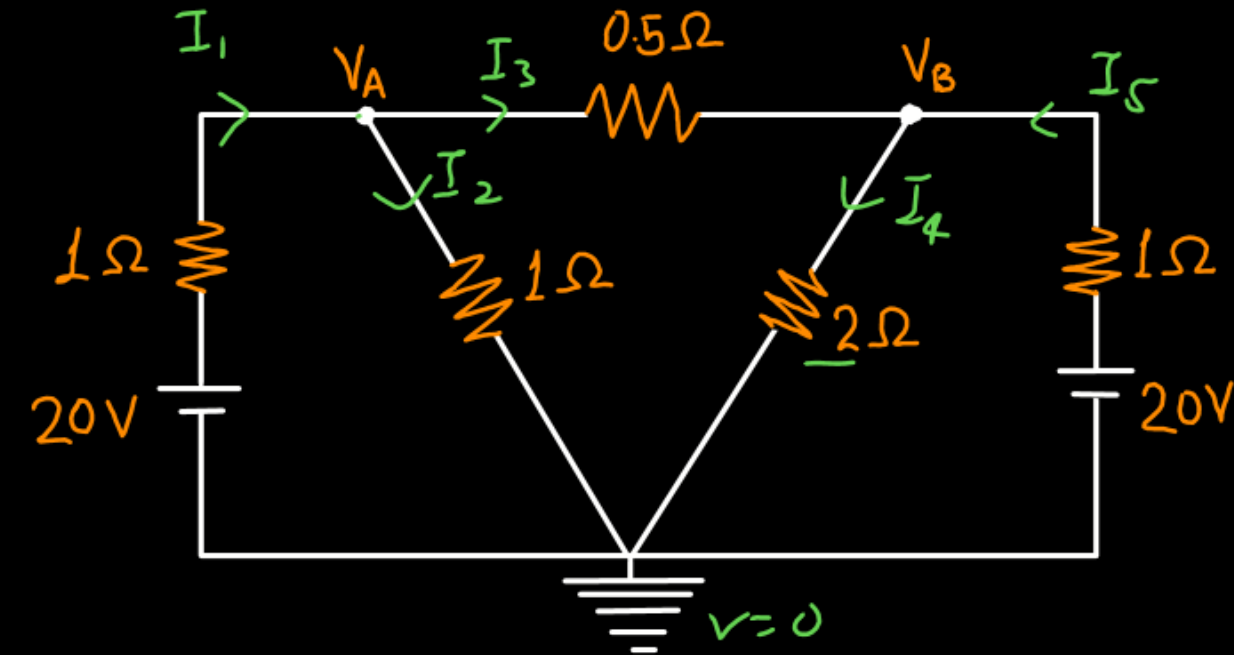
Solve Eqn ① & ②

$$V_A = 11 \text{ V}$$

$$V_B = 12 \text{ V}$$

Ans

$$I_4 = \frac{V_B}{2} = \frac{12}{2} = 6 \text{ amp}$$



Q.9 Calculate the current across 6Ω resistor in the following circuit using: (AKTU 2023-24)

~~Mesh Analysis~~ $I_1 = \frac{(18 - V_A)}{3}$, $I_2 = \frac{V_A}{6}$, $I_3 = \frac{12 - V_A}{7}$

Nodal Analysis

Apply KCL at Node

$$I_1 + I_3 - I_2 = 0$$

$$\left(\frac{18 - V_A}{3}\right) + \left(\frac{12 - V_A}{7}\right) - \frac{V_A}{6} = 0$$

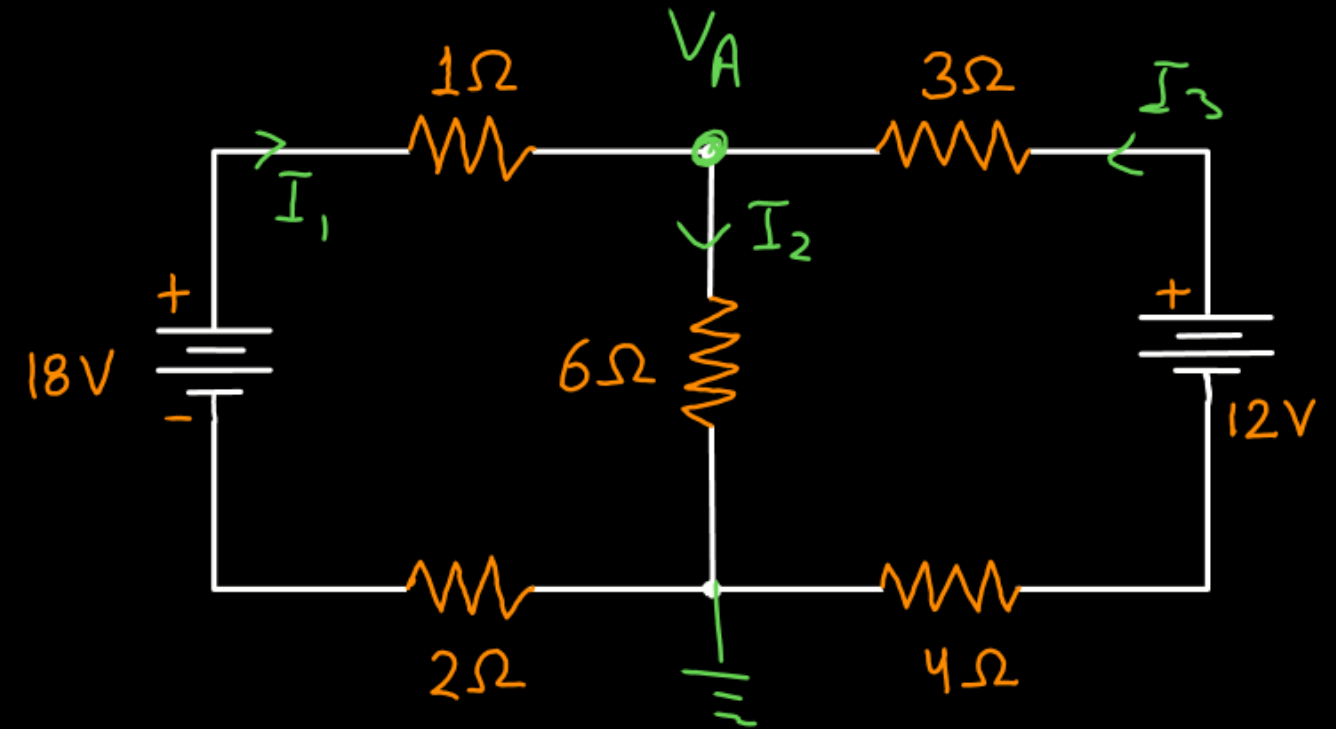
$$\frac{18}{3} - \frac{V_A}{3} + \frac{12}{7} - \frac{V_A}{7} - \frac{V_A}{6} = 0$$

$$7.71 - V_A \left(\frac{1}{3} + \frac{1}{7} + \frac{1}{6}\right) = 0$$

$$7.71 = 0.64 V_A$$

$$V_A = 11.99 \approx 12V$$

$$I_2 = \frac{V_A}{6} = \frac{12}{6} = 2 \text{ amp}$$



Q.10 Find the value of current in 8Ω resistance using Nodal Analysis

$$I_1 = \frac{15 - V_B}{20}, \quad I_2 = \frac{15 - V_B}{8}, \quad I_3 = \frac{V_B - 0 + 25}{20} = \frac{V_B + 25}{20}$$

Apply KCL at Node

$$I_1 + I_2 - I_3 = 0$$

$$\frac{15 - V_B}{20} + \frac{15 - V_B}{8} - \frac{V_B + 25}{20} = 0$$

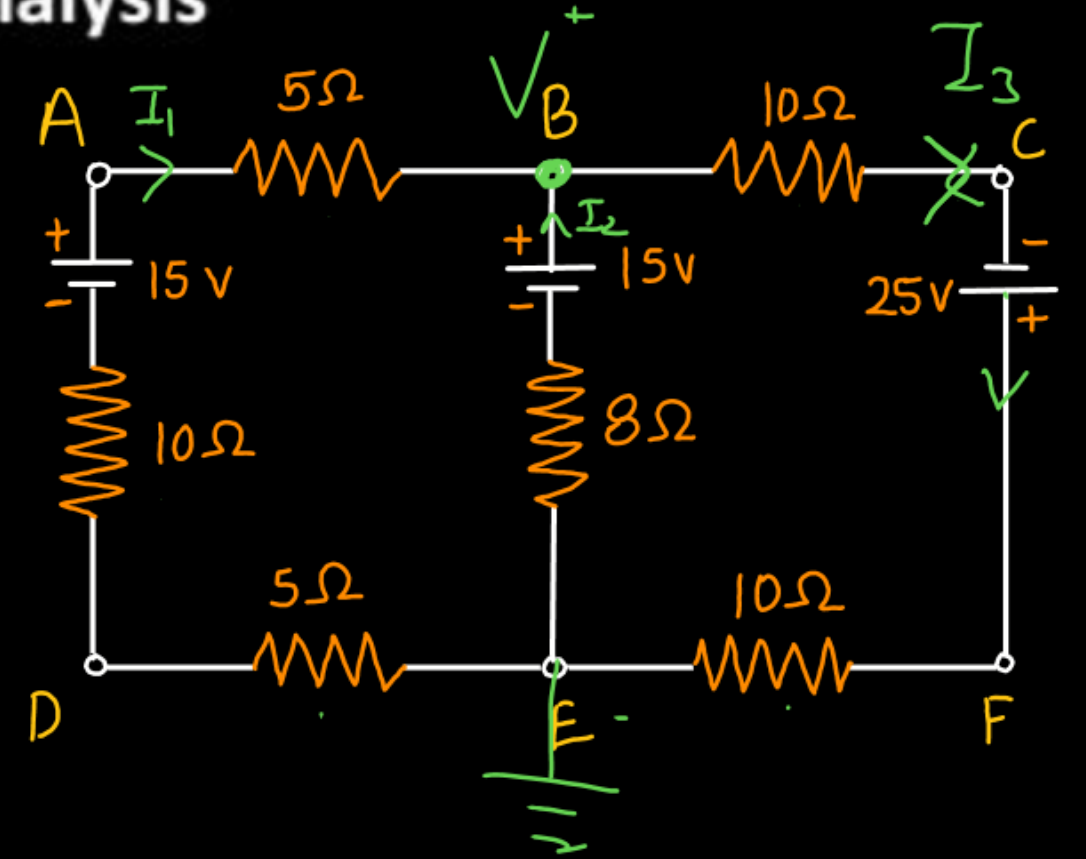
$$\left(\frac{15}{20} + \frac{15}{8} - \frac{25}{20} \right) - V_B \left(\frac{1}{20} + \frac{1}{8} + \frac{1}{20} \right) = 0$$

$$1.375 - V_B (0.225) = 0$$

$$V_B = \frac{1.375}{0.225} = 6.11 \text{ Volt}$$

$$I_{8\Omega} = I_2 = \frac{15 - 6.11}{8} = 1.11 \text{ Amp}$$

Ans



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Thank You